Exercise: Coordinates, Axes, and Scales

Overview

This exercises provides some practice cleaning up axes for data visualizations and is designed to be completed in 60 minutes or fewer. Although you should have time to complete the parts, if you are unable, *as with all exercises, you are encouraged to complete it outside of class time* so that you are able to incorporate your experiences and knowledge into future exercises. You may need to consult course reading materials located at the course site as some elements may not have been covered in the basic content contained in associated videos. If you would like to acquire extra credit for your work, you can send me a knit html file of what you have accomplished.

This exercise utilizes your understanding of `{dplyr}` functions like `select()`, `filter()`, `mutate()`, and `summarize()` and `{ggplot2}` functions like `ggplot()`, `geom_point()`, `geom_bar()`, and as `aes()` and `{scales}` functions. Collaborate with others and reference readings if necessary.

This exercise focuses on:

- Replicating data visualizations
- Adjusting axes scales (e.g., labels, breaks)
- Using functions in `{scales}`
- Working with color palettes using `{monochromeR}`

Data Set

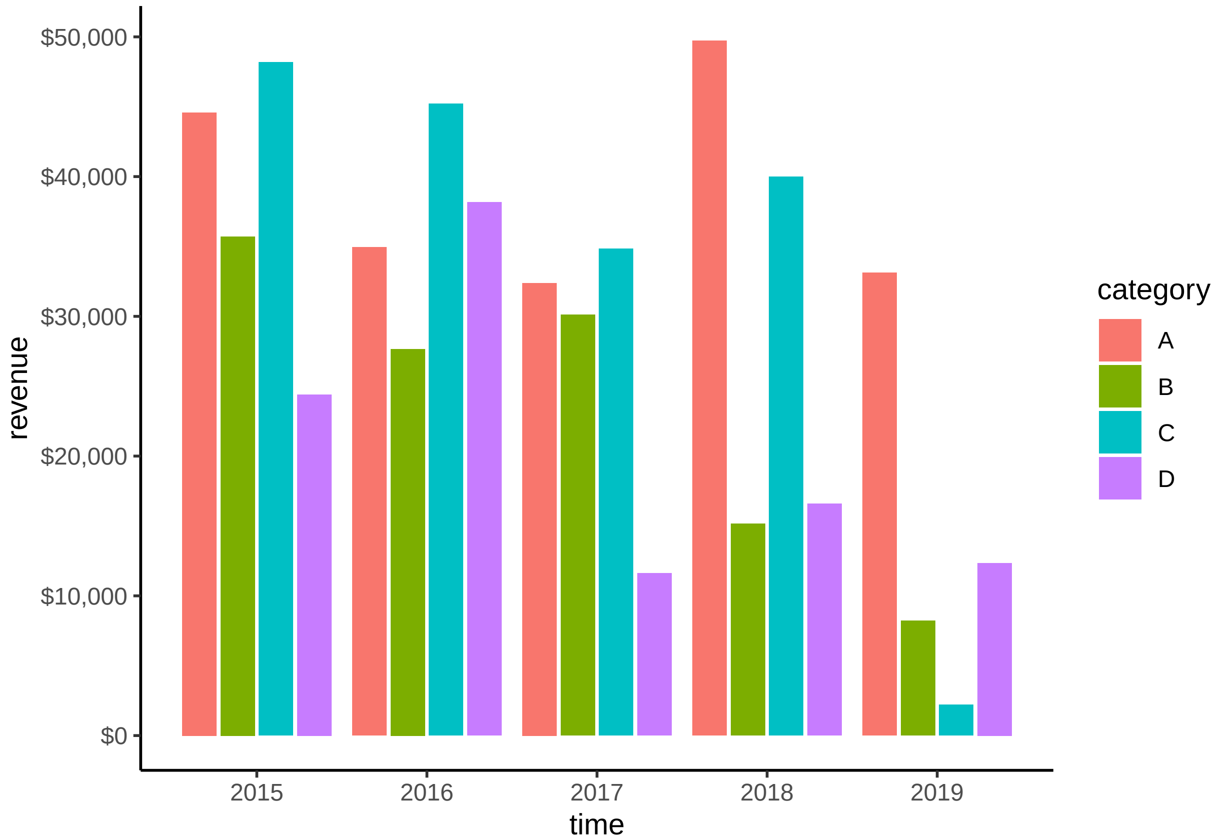
In this exercise, you will create several plots using `{ggplot2}` and modify axis labels and breaks using the `{scales}` library. Use the provided data set to complete the problems.

```r
set.seed(123)

data <- data.frame(
  category = rep(c("A", "B", "C", "D"), each = 5),
  proportion = runif(20, 0.01, 1),
  revenue = round(runif(20, 1000, 50000), 2),
  time = rep(2015:2019, 4)
)
```

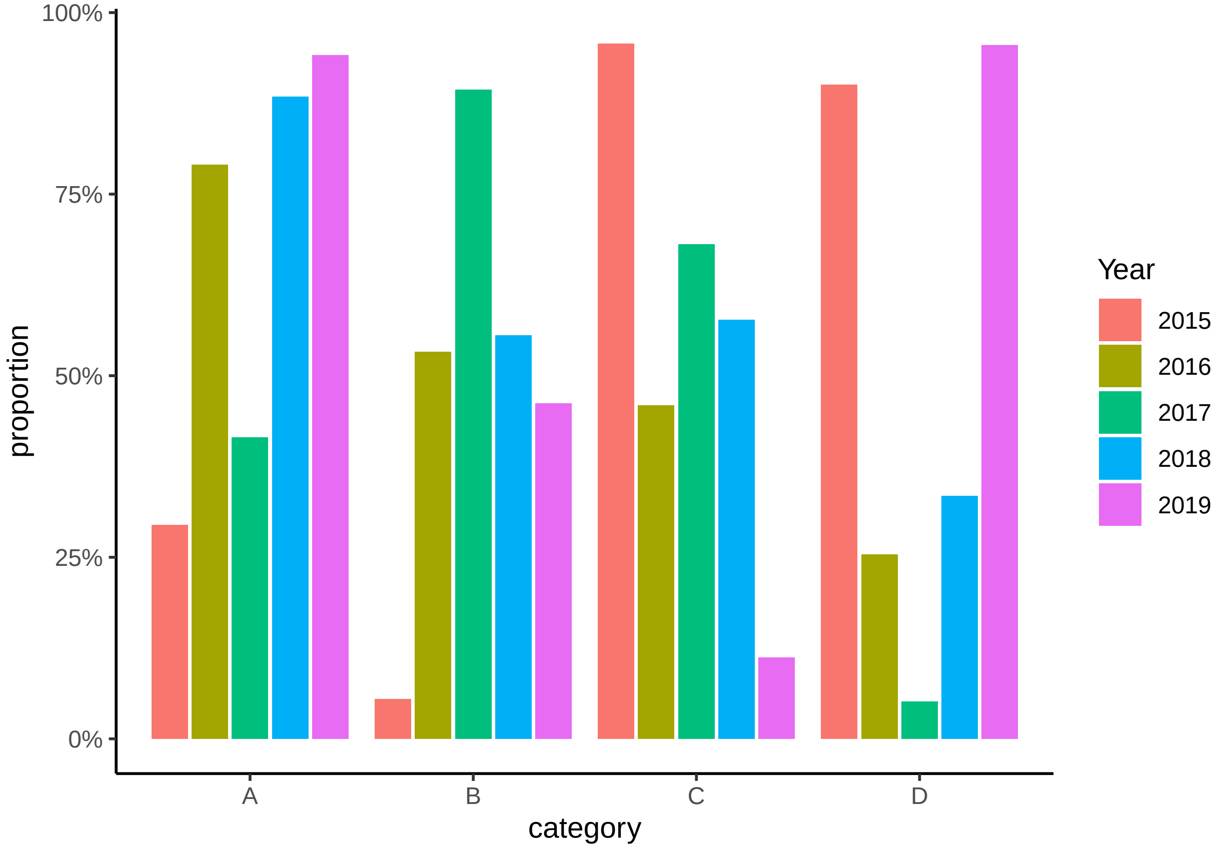
Problem 1

Review `{scales}` functions so that you can display your axis as *dollars* and replicate this plot. But also consider the data-to-ink ratio and ask yourself whether all the 0's are needed for such plots.



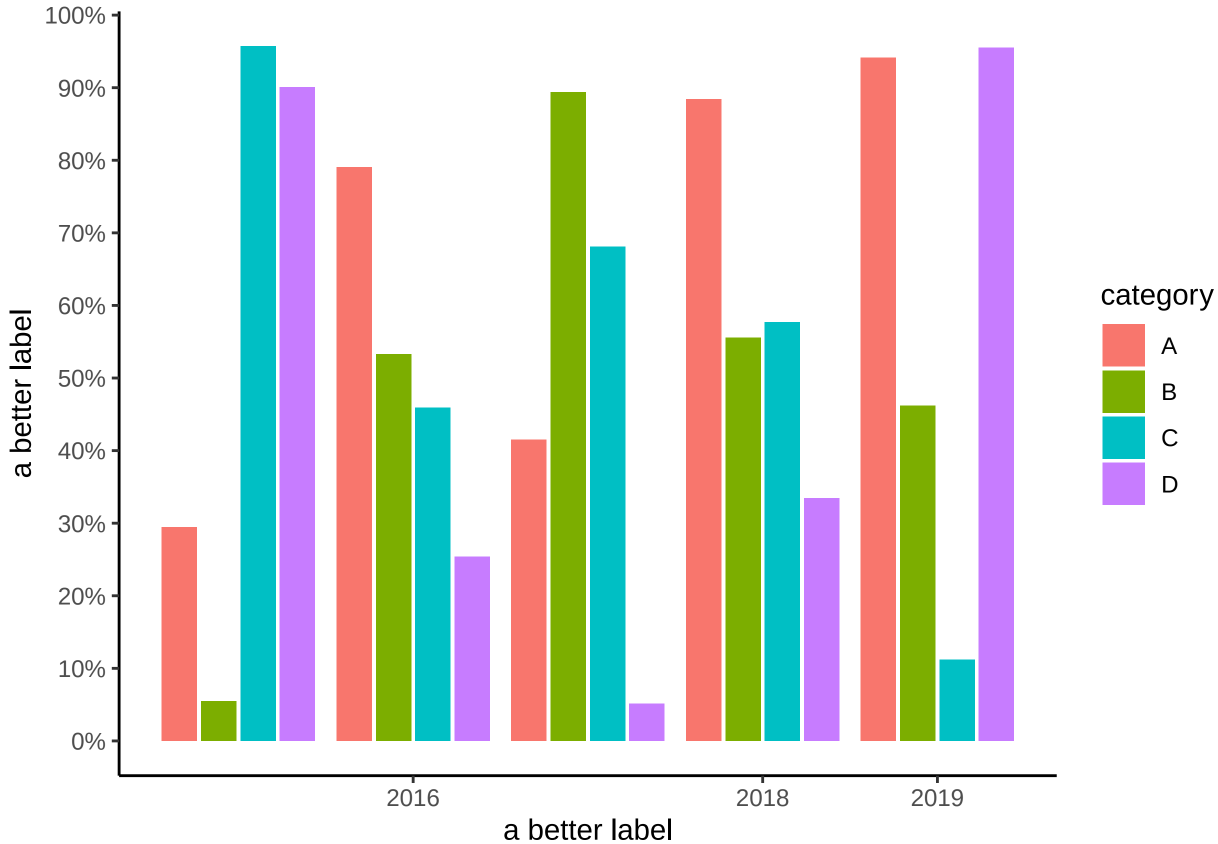
Problem 2

Review {scales} functions so that you can display your axis as *percentages* and replicate this plot.



Problem 3

Replicate the following plot, making sure to limit the x-axis to display only the years 2016, 2018, and 2019 by changing the scale. This plot may not represent what you might desire to create but it nevertheless provides experience understanding how scale function work. Consider also labeling your axis in a way that make senses based on your modification.



Problem 4

Create a custom color palette of discrete colors using `{monochromeR}` in order to change the colors of either `category` or `year` for one of your plots.

Problem 5 (Bonus): Mutating a Custom Color Palette

Having created a custom palette, add a column to your data frame (e.g., `cat_color`) that contains the custom colors that would correspond to a variable that you would use for adding color to the plot. Consider what `{dplyr}` function you might use to create variables with more than two unique elements. These color values in the variable can now be referenced by their `"identity"` when applying a color. Use `scale_*_identity()` to color your bars or points of your plot. This problem will provide you with practice with using colors if they are in a data frame already. When colors are builty into the data frame, there is no question about the colors used.